

9. (a) The Earth's early atmosphere around 4 000 million years ago contained mainly carbon dioxide and water vapour produced by volcanoes.

(i) Explain why the large percentage of water vapour in the Earth's atmosphere decreased over geological time. [2]

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(ii) Give **two** reasons why the percentage of carbon dioxide in the Earth's atmosphere has decreased over geological time. [2]

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(b) During the last 250 years the percentage of carbon dioxide in the Earth's atmosphere has increased from 0.03 % to 0.04 %. This has led to increased global warming. Give **one** reason for this increase and explain why global warming is a cause for concern. [2]

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(c) Ammonia present in the Earth's early atmosphere reacted with oxygen to produce nitrogen and water vapour. Complete the balancing of the symbol equation for this reaction. [1]



END OF PAPER

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